

Date	9 July 2026	Project Name	Heart Disease Analysis
Team ID	—	Maximum Marks	4 Marks

PROJECT DEVELOPMENT PHASE

Data Preprocessing & Dashboard Development Pipeline

Team Leader: Mounika Sanapathi

Team Members: Thakasi Balaji, Kale Sai Praneeth, Navyasri Gonnuri, Shanthi Kumari Challa

Data Preprocessing Steps

Step 1: Import the Dataset

Imported the clinical Heart Disease Analysis dataset into Tableau Public. Verified that all records and metrics (Patient ID, Age Group, Chest Pain Type, Resting Blood Pressure, Serum Cholesterol, Fasting Blood Sugar, Resting ECG, Max Heart Rate Achieved, Patient Risk Segment, Hospital Branch Location) were parsed successfully and confirmed dataset readiness for pipeline alignment.

Step 2: Check for Missing Values

Examined the diagnostic variables for null, empty, or unrecorded clinical entries. Checked that critical attributes like Cholesterol, Blood Pressure, Chest Pain Classification, Max Heart Rate, and Branch Location held valid entries. Handled any blank parameters appropriately to maintain model precision.

Step 3: Remove Duplicate Records

Checked the repository for duplicate case tracking records using the distinct Patient ID field. Erased recurring logs to avoid statistical inflation and misaligned diagnostic weights, ensuring each profile was counted exactly once.

Step 4: Validate Data Types

Configured proper format maps and schemas for all physiological fields:

- Patient ID → String / Text Identification
- Serum Cholesterol → Number
- Resting Blood Pressure → Number
- ST Depression Metric → Decimal Number
- Max Heart Rate Achieved → Number
- Chest Pain Type → Categorical Variable
- Resting ECG Results → Categorical Variable
- Patient Risk Segment → Categorical Variable
- Hospital Branch Location → Geographic / String Mapping

Step 5: Verify Data Consistency

Audited fields for unrealistic or invalid entries such as negative blood pressure metrics, impossible cholesterol counts, or out-of-bounds heart rate parameters. Standardized chest pain descriptions, location titles, and category structures to align categorical fields perfectly before visualization.

Step 6: Prepare Calculated Fields

Created the calculated fields and key clinical measures required for aggregated dashboard KPIs:

- Total Monitored Cohorts
- Average Patient Cholesterol Levels
- Average Resting Blood Pressure Baseline
- Max Heart Rate Distribution Matrix
- Risk Density Cross Analysis

Step 7: Final Clean Dataset

Verified that the cleaned diagnostic dataset was comprehensive, complete, and fully structured for production deployment. Finalized the architecture layers to generate business intelligence visualizations and live stories inside Tableau Public.

Business Questions with Visualization Mapping

S.No.	Business / Clinical Question	Target Visualization Type
1	How many patient records are available for analysis after data preprocessing?	Total Case Count (KPI Card)
2	What is the overall average cholesterol score across the patient demographic?	Average Cholesterol Level (KPI Card)
3	Which chest pain type category correlates with the highest number of high-risk diagnoses?	Chest Pain Risk Correlation Map
4	How do patient volumes and conditions vary across different hospital branches?	Branch-wise Patient Volume Matrix
5	What is the average diagnostic confidence rate across risk segments?	Diagnostic Evaluation Analysis
6	How does the maximum achieved heart rate vary among different patient risk tiers?	Max Heart Rate Tier Distribution
7	What are the clinical distribution trends across resting ECG types?	Resting ECG Category Matrix
8	How does cardiovascular patient risk behavior differ across various age groups?	Patient Age Group Distribution Map

Visualizations & Dashboard Core Measures

Below are the structural metrics validated after data preprocessing and dashboard assembly:

TOTAL PATIENTS ANALYZED 50,000	AVERAGE CHOLESTEROL (MG/DL) 247.73	TOTAL CLINICAL RECORDS MAP 27,386,555.00
--	--	---

Max Heart Rate Achieving Distribution: A uniform tracking chart breaks down recorded occurrences into heart rate bins ranging across standard test intervals, displaying optimal data density in the middle bands with tail reductions at maximum cardiovascular limits.

Patient Distribution Matrix: Tracks diagnostic occurrences filtered by city and temporal shifts (e.g., *Mumbai, Pune, Bangalore, Delhi, Hyderabad, Chandigarh* mapped against morning, afternoon, and evening check-ins), enabling staff to track peak traffic flows and adjust operational capacity accordingly.